Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Run Ablation Study
2. Abbreviated Title	Ablation Study
3. Use Case Id	UC-08
4. Actors	Clinician (Primary), ML Engine (Supporting)
5. Description	The Clinician runs ablation experiments to evaluate the contribution of individual model components (CNN branch, LSTM branch, Attention mechanism) to the overall prediction performance.
5.1. Pre Conditions	1. A fully trained CNN-LSTM-Attention model exists. 2. Test dataset is available for evaluation.
5.2. Task Sequence	1. Clinician selects components to ablate (e.g., remove Attention layer). 2. System creates a modified model variant with the component disabled. 3. ML Engine evaluates the modified model on the test set. 4. System records performance metrics for the ablated variant. 5. Steps 2-4 repeat for each ablation configuration. 6. System generates a comparison table/chart of all variants.
5.3. Post Conditions	1. Comparative metrics (with/without each component) are generated. 2. Component contribution to overall accuracy is quantified.
6. Modification History	Date: September 11, 2026
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